Government Debt and Budget Deficits

Chapter 19

Econ 313 – Intermediate Macroeconomics

University of Tennessee, Knoxville

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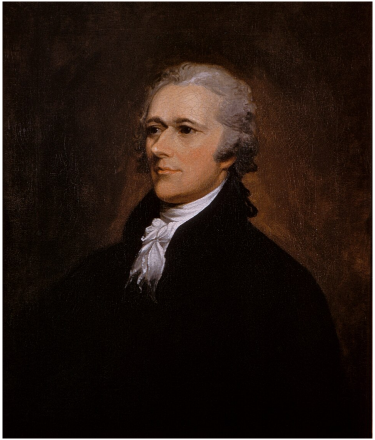
In This Chapter, You Will Learn:

- About the size of the U.S. government's debt and how it compares to that of other countries
- About problems measuring the budget deficit
- About the traditional and Ricardian views of the government debt
- About other perspectives on government debt

Gross debt of general government (2021)

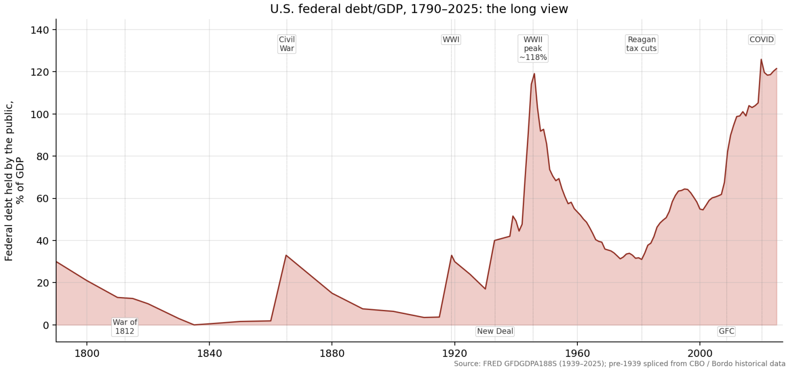
Country	**Debt as a % of GDP**
Japan	256
Greece	224
Italy	173
United States	148
Portugal	144
Spain	143
United Kingdom	143
France	138
Canada	134
Belgium	129
Australia	84
Germany	77
Netherlands	67
Switzerland	-42



Alexander Hamilton, first U.S. Treasury Secretary. The U.S. federal government has carried debt every year since Hamilton refunded the Revolutionary War debt in 1790.

J. Trumbull, 1806 / Wikimedia, PD

U.S. federal debt/GDP, 1790–2025 (the long view)



Debt was near-zero under Jackson (1835), spiked in every major war, and reached today's level (120%) only twice — WWII and COVID.

Source: FRED `GFDGDPA188S`; pre-1939 spliced from CBO / Bordo historical data.

Liberty Bond, WWI.
How wars were financed.

LoC / Wikimedia, PD

The U.S. experience in recent years, part 1

Early 1980s through the early 1990s:

- debt-to-GDP ratio: 25.5% in 1980, 48.9% in 1993
- due to Reagan tax cuts and increases in defense spending and entitlements

Early 1990s through 2000:

- \$290 billion deficit in 1992, \$236 billion surplus in 2000
- debt-to-GDP ratio fell to 32.5% in 2000
- due to rapid growth, stock market boom, tax hikes

Richard Nixon, 1971. Closed the gold window 15 Aug 1971, ending Bretton Woods and opening the fiat-money era of large deficits.

Wikimedia, PD

The U.S. experience in recent years, part 2

The 2008–2009 recession and its aftermath:

- huge spending increases; bailouts of banks (TARP) and auto industry, and a stimulus package (ARRA).
- TARP: 431 billion, ARRA: 787 billion.
- Weak recovery kept tax revenues low

2017: Tax Cut and Jobs Act (TCJA)

- Large tax cut is estimated to increased debt by ∼2 trillion over 10 years (CBO)

Covid-19 recession:

- Multiple relief bills passed to help unemployed people and the economy
- CARES: 2.2 trillion, December 2020 bill: 900 billion. ARP: 1.9 trillion

The troubling long-term fiscal outlook

- The U.S. population is aging.
- Healthcare costs are rising.
- Spending on entitlements like Social Security and Medicare is growing.
- Deficits and the debt are projected to significantly increase...

Problems measuring the deficit

1. Inflation
2. Capital assets
3. Uncounted liabilities
4. The business cycle

MEASUREMENT PROBLEM 1: Inflation, part 1

- Suppose the real debt is constant, which implies a zero real deficit.
- In this case, the nominal debt D grows at the rate of inflation:

$$\Delta D/D = \pi \quad \text{or} \quad \Delta D = \pi D$$

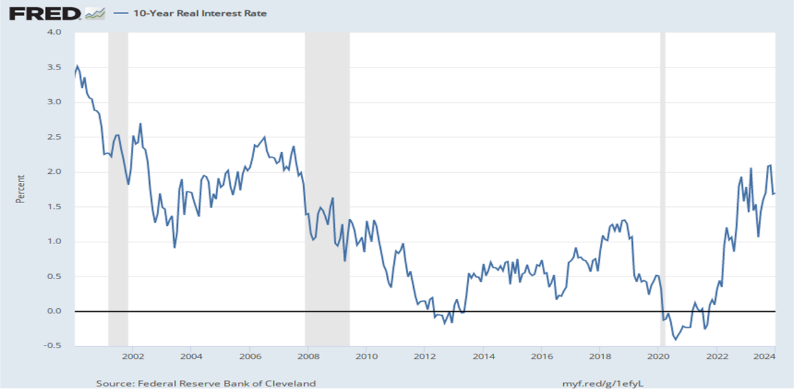
- The reported deficit (nominal) is πD even though the real deficit is zero.
- Hence, we subtract πD from the reported deficit to correct for inflation.

MEASUREMENT PROBLEM 1: Inflation, part 2

- Correcting the deficit for inflation can make a huge difference, especially when inflation is high.
- What matters is the *real* debt, and therefore, the *real* interest rate on government debt.
- For example: In 1979,
  - nominal deficit = \$28 billion
  - inflation = 8.6 percent
  - debt = \$495 billion
  - $\pi D = 0.086 \times \$495$ billion $= \$43$ billion
  - real deficit = \$28 billion − \$43 billion = \$15 billion surplus

MEASUREMENT PROBLEM 1: Inflation, part 3

- 10-Year real interest rate



MEASUREMENT PROBLEM 2: Capital assets, part 1

- Currently, deficit = change in debt
- Better, **capital budgeting**:
  deficit = (change in debt) − (change in assets)
- For example: Suppose the government sells an office building and uses the proceeds to pay down the debt.
  - under current system, deficit would fall
  - under capital budgeting, deficit unchanged, because fall in debt is offset by a fall in assets
- Problem with capital budgeting: determining which government expenditures count as capital expenditures

MEASUREMENT PROBLEM 2: Capital assets, part 2

- There have been recent proposals for massive government spending on fixing and expanding U.S. infrastructure
  - Crumbling bridges, tunnels, and water pipes
  - Green New Deal:
    - Would have the government build clean-energy public transit and expand solar and wind energy, among other items
- These would increase *assets*, while increasing the *deficit*. Capital budgeting puts the plan in a different light.

MEASUREMENT PROBLEM 3: Uncounted liabilities

- The current measure of the deficit omits important liabilities of the government:
  - future pension payments owed to current government workers
  - future Social Security payments
  - contingent liabilities, such as covering federally insured deposits when banks fail
- (It is difficult to attach a dollar value to contingent liabilities due to the inherent uncertainty.)

MEASUREMENT PROBLEM 4: The business cycle, part 1

- The deficit varies over the business cycle due to automatic stabilizers (unemployment insurance, the income tax system).
- These are not measurement errors but do make it harder to judge fiscal policy stance.
  - For example: Is an observed increase in the deficit due to a downturn or an expansionary shift in fiscal policy?

MEASUREMENT PROBLEM 4: The business cycle, part 2

Solution: **cyclically adjusted budget deficit** (aka full-employment deficit) based on estimates of what government spending and revenues would be if the economy were at the natural rates of output and unemployment.

The bottom line

We must exercise care when interpreting the reported deficit figures.

Is government debt really a problem?

Consider a tax cut with corresponding increase in the government debt.

Two viewpoints:

1. Traditional view
2. Ricardian view

The traditional view

- Short run: $\uparrow Y$, $\downarrow u$
- Long run:
  - Y and u back at their natural rates
  - closed economy: $\uparrow r$, $\downarrow I$
  - open economy: $\uparrow \varepsilon$, $\downarrow NX$
    (or higher trade deficit)
- Very long run:
  - slower growth until the economy reaches a new steady state with lower income per capita

The Ricardian view

- Due to David Ricardo (1820), advanced more recently by Robert Barro
- David Ricardo famously did not believe the idea that now has his name
- According to **Ricardian equivalence**, a debt-financed tax cut has <u>no effect</u> on consumption, national saving, the real interest rate, investment, net exports, or real GDP, <u>even in the short run</u>.

The logic of Ricardian equivalence

- Consumers are forward-looking, know that a debt-financed tax cut today implies an increase in future taxes that is equal—in present value—to the tax cut.
- The tax cut does not make consumers better off, so they do not increase consumption spending.
- Instead, they save the full tax cut in order to repay the future tax liability.
- Result: Private saving rises by the amount public saving falls, leaving national saving unchanged.

Numerical example

Problems with Ricardian equivalence

- **Myopia**: Not all consumers think so far ahead; some see the tax cut as a windfall.
- **Borrowing constraints**: Some consumers cannot borrow enough to achieve their optimal consumption, so they spend a tax cut.
- **Future generations**: If consumers expect that the burden of repaying a tax cut will fall on future generations, then a tax cut now makes them feel better off, so they increase spending.

Evidence against Ricardian equivalence? Part 1

- During the summer of 2001, \$300 to \$600 "tax rebates" (direct checks that served as a one-time, lump-sum tax cut) were mailed to people.
- The tax rebate increased the deficit (overall bill increased the deficit by 1.2 trillion from 2001 to 2011).
- Consumers spent roughly *two-thirds* of their rebate in three months after receiving the check.

Evidence against Ricardian equivalence? Part 2

- Ricardian equivalence would have suggested consumers save the check for future tax increases.
- Consistent with the borrowing constraint explanation, spending was even greater for consumers with low income and low wealth.
- Similar studies of the 2008 stimulus payment found that consumers spent up to 90 percent of the payment.

OTHER PERSPECTIVES: Balanced budgets versus optimal fiscal policy

- Some politicians have proposed amending the U.S. Constitution to require a balanced federal government budget every year.
- Many economists reject this proposal, arguing that the deficit should be used to:
  - stabilize output and employment
  - smooth taxes in the face of fluctuating income
  - redistribute income across generations, when appropriate

OTHER PERSPECTIVES: Fiscal effects on monetary policy

- Government deficits may be financed by printing money.
- A high government debt may be an incentive for policymakers to create inflation (to reduce real value of debt at expense of bond holders).

Fortunately:

- little evidence that the link between fiscal and monetary policy is important
- most governments know the folly of creating inflation
- most central banks have at least some political independence from fiscal policymakers

OTHER PERSPECTIVES: Debt and politics

"Fiscal policy is made not by benevolent, all-knowing angels but by government officials engaged in an imperfect political process."

—N. Gregory Mankiw

- Some do not trust policymakers with deficit spending. They argue that:
  - policymakers do not worry about true costs of their spending, since the burden falls on future taxpayers.
  - since future taxpayers cannot participate in the decision process, their interests may not be taken into account.

OTHER PERSPECTIVES: International dimensions

- Government budget deficits can lead to trade deficits, which must be financed by borrowing from abroad.
- Large government debt may increase the risk of capital flight, as foreign investors may perceive a greater risk of default.
- Large debt may reduce a country's political clout in international affairs.

Chapter Summary, Part 1

- Relative to GDP, the U.S. government's debt is moderate compared to that of other countries.
- Standard figures on the deficit are imperfect measures of fiscal policy because they:
  - are not corrected for inflation.
  - do not account for changes in government assets.
  - omit some liabilities (such as future pension payments to current workers).
  - do not account for the effects of business cycles.

Chapter Summary, Part 2

- In the traditional view, a debt-financed tax cut increases consumption and reduces national saving. In a closed economy, this leads to higher interest rates, lower investment, and a lower long-run standard of living. In an open economy, it causes an exchange rate appreciation and a fall in net exports.
- The Ricardian view holds that debt-financed tax cuts do not affect consumption or national saving and therefore do not affect interest rates, investment, or net exports.

Chapter Summary, Part 3

- Most economists oppose a strict balanced budget rule, as it would hinder the use of fiscal policy to stabilize output, smooth taxes, or redistribute the tax burden across generations.
- Government debt can have other effects:
  - It may lead to inflation.
  - Politicians can shift the burden of taxes from current to future generations.
  - It may reduce the country's political clout in international affairs or scare foreign investors into pulling their capital out of the country.

Appendix: Ricardian equivalence — a two-period example

A consumer lives two periods, receives income Y_1, Y_2, pays taxes T_1, T_2. Real interest rate $r = 0$ (simplifies arithmetic).

Baseline. $Y_1 = 100,\ Y_2 = 100,\ T_1 = 20,\ T_2 = 20.$ Lifetime resources =

$$(100 - 20) + (100 - 20) = 160.$$

Consumer smooths $C_1 = C_2 = 80.$

Policy: debt-financed tax cut. Government cuts T_1 by 10 (to 10), finances via bond that must be repaid with interest next period. Since $r = 0$: T_2 rises to 30.

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Appendix — the Ricardian consumer's response

Consumer's after-tax income:

$$\begin{aligned} &\text{Period 1:} \quad Y_1 - T_1 = 100 - 10 = 90 \quad \text{(up by 10)} \\ &\text{Period 2:} \quad Y_2 - T_2 = 100 - 30 = 70 \quad \text{(down by 10)} \end{aligned}$$

Lifetime resources: $90 + 70 = 160$ — *exactly the same as before.*
Consumer still wants smooth consumption: $C_1 = C_2 = 80$. So the tax cut's $10 goes entirely into *private saving*:

$$S_1^{\text{priv}} = Y_1 - T_1 - C_1 = 90 - 80 = 10.$$

Public saving drops from 0 (in baseline) to -10. National saving:

$$S_1^{\text{nat}} = S_1^{\text{priv}} + S_1^{\text{pub}} = 10 + (-10) = 0.$$

Result: $\Delta C = 0$, $\Delta S^{\text{nat}} = 0$, $\Delta r = 0$, $\Delta I = 0$.
The tax cut is *neutral*.

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Appendix — why Ricardian equivalence often fails

The above requires strong assumptions. Any one of these breaks it:

- **Myopia.** Consumer ignores T_2: sees a \$10 windfall and spends part of it. $\Delta C > 0$, $\Delta S^{\text{nat}} < 0$.
- **Borrowing constraints.** Cannot borrow against future income: the \$10 relaxes today's constraint and gets spent.
- **Finite lives.** Future taxes fall on *other people* (children, immigrants). Ricardo only holds if generations act as one dynasty (Barro 1974).
- **Distortionary taxes.** Taxes distort labor and saving, so *when* they are levied changes real behavior.

Empirical evidence is mixed. 2001, 2008, 2020 U.S. tax rebates: households consumed about 20–40% within a quarter — not zero, not one. Reality is between traditional and Ricardian.

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